

What is a Human Movement Assessment (HMA)?

Human Movement Assessments are a systematic approach to identifying opportunities to improve human mobility and stability.

What we know:

- ❖ Deficits in human movement (mobility, stability, coordination/balance) contribute to musculoskeletal (MSK) injury.
- ❖ Successfully identifying opportunities to improve human movement allows ATI Worksite Solutions (AWS) to intentionally mitigate injury risk.
- ❖ Human Movement Assessment is a tool that create mutual recognition of movement deficits; therefore, open the opportunity to address opportunities proactively and precisely.

Why these 5 movements?

- ❖ **LUNGE:** Functional, assesses balance; trunk extension; hip extension (often lacking); LE dissociation; trunk/hip dissociation; hip, knee, ankle flexion.
- ❖ **SINGLE LEG DIP:** Functional; assesses LE stability; assesses balance; assesses fear/avoidance.
- ❖ **SHOULDER REACH:** Functional; assesses thoracic extension (huge); assesses shoulder flexion, extension, IR, ER; clears shoulder for current injury.
- ❖ **TRUNK ROTATION:** Functional; assesses lumbar rotation, assesses thoracic rotation and unilateral extension, assesses breath control (stress).
- ❖ **CERVICAL ROTATION:** functional; assesses cervical rotation, assesses chin tuck.

Recommendations Packet

The provided recommendation templates serve as essential resources for you, the providers, in managing the musculoskeletal (MSK) deficits identified in the HMA. It is important to recognize that each employee presents a unique set of circumstances and needs. Therefore, while these templates offer valuable guidance, they are not exhaustive. Providers are encouraged to customize and modify the recommendations based on their clinical expertise and the specific requirements of each individual employee.

As the onsite MSK expert, your professional judgement is crucial in identifying additional exercises that may not be included in the templates. This tailored approach ensures that the employees onsite receive the most effective and appropriate care, optimizing their overall well-being. Your ability to adapt these recommendations will not only enhance outcomes but also foster a more personalized employee experience.

LEVATOR SCAPULAE STRETCH



Start by sitting or standing up straight with your head and shoulders relaxed. Slowly tilt your head to one side aiming your ear at your shoulder and hold without shrugging your shoulders. Slowly transition to the other side. Repeat as directed by your provider.

CERVICAL SIDE BENDING



Start by sitting or standing up straight with your head and shoulders relaxed. Slowly tilt your head to one side aiming your ear at your shoulder and hold without shrugging your shoulders. Slowly transition to the other side. Repeat as directed by your provider.

STERNOCLEIOMASTOID STRETCH



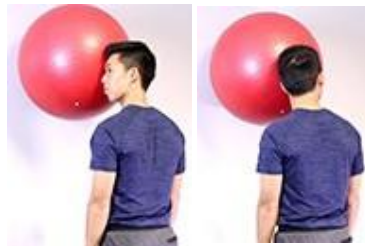
Start by sitting up straight in a chair. Place both hands on your collarbone on the side you want to stretch and gently pull down to maintain your shoulder position. Slowly tilt your chin up and tilt your head away from your hands. Hold as directed by your provider.

ISOMETRIC CERVICAL RETRACTION



Start by standing with your back against a wall and a towel behind the back of your head. Attempt to extend your head back against the wall and hold the extension. You will not actually move because the wall is blocking this motion.

CERVICAL ROTATION WITH BALL



Start by standing with a wall in front of you and an exercise ball in your hands. Place the ball on the wall in front of your face and hold the ball against the wall with your forehead. Your eyes should be looking straight ahead. Slowly rotate your head all the way to one side, allowing the ball to roll along your head. Then, rotate it all the way to the other side.

TREAD THE NEEDLE WITH EXTENSION



Start on your hands and knees. Brace your abdominals and slowly lift one arm straight out to the side with your palm facing the floor. Slowly bring your arm back down.

ISOMETRIC CERVICAL ROTATION



Start by placing your fingers of one hand on the side of your forehead near your temples. Attempt to rotate your head in the direction of your fingers but resist this motion with your hand so that your head does not actually move. Your head should remain in a neutral position the entire time. Hold and then repeat on the other side.

CHIN TUCK



Start by looking straight ahead with your head in a neutral position. Place your finger on the tip of your chin and gently press inward so your chin tucks in and your head retracts back. Do not move anything except your head.

ISOMETRIC CERVICAL SIDE BENDING



Start by looking straight ahead with your head in a neutral position. Place two fingers from one hand on your temple and attempt to push your head to the side. Resist this motion with your head using your neck muscles. Your head should not actually move.

SINGLE LEG BALANCE



Start by standing up straight with your feet close together. Slowly raise one foot off the ground and bring your arms up and out to your sides to aid with balance. Make sure to brace your abdominals. Perform with each leg

TANDEM BALANCE



Start by placing the foam pad down on the ground. Step onto the foam pad so that one heel is in the corner of the pad and the other heel is right in front of the toes of your other foot. Repeat with the other leg in front

SINGLE LEG TOE TOUCHES



Start by standing on one leg with the other leg slightly off the ground. Slowly tap your raised foot out to the side and then across your body to the other side while maintaining balance.

FORWARD WALK WITH HEAD TURN



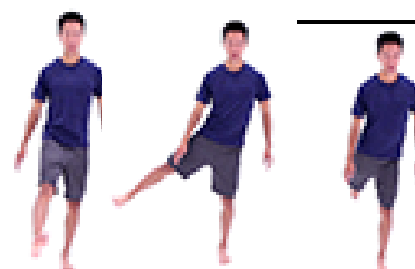
Start by standing up straight. Start walking forward starting with your left foot. Move your head up diagonally and then down diagonally toward the opposite side as you walk forward for five steps. Turn around and repeat leading with the right foot

MONSTER WALK



Raise both arms straight out in front of you to about shoulder height with your palms facing down. Kick one leg straight up toward your palms and then place it back down slightly forward. Then, repeat with your other leg. Continue walking forward in this fashion for ten steps.

SINGLE LEG CLOCK BALANCE



Start by standing up straight and raising one foot slightly off the ground. Move your raised, straight leg forward and back to neutral. Then out to the side and back and then backward and back to neutral. Repeat this three times with each leg

SINGLE LEG SIT TO STAND



Start seated tall in a chair, hands by your sides, feet flat on the floor. Lift one foot off the ground, and stand all the way up, keeping your non-working leg lifted off the ground. Slowly lower yourself back down to the chair.

SINGLE LEG FORWARD BALANCE



Start by standing up straight. Raise both arms straight up over your head and kick one leg straight back behind you as you bend forward at your hips like a hinge until your back leg, torso, and arms parallel the floor.

CROSS OVER JUMPS



Start by standing up straight. Jump to the left and land softly on your left leg. As you land, swing your right leg slightly behind your left and tap your toes to the floor. Repeat by jumping onto your other foot.

HALF KNEELING CALF STRETCH



Start on one knee with the leg you want to stretch in front. Place your hands on your front knee and slowly press forward. Push your knee forward without lifting your heel until you feel a stretch in the back of your lower leg. Hold as directed by your provider.

SEATED PIRIFORMIS STRETCH



Start on one knee with the leg you want to stretch in front. Place your hands on your front knee and slowly press forward. Push your knee forward without lifting your heel until you feel a stretch in the back of your lower leg. Hold as directed by your provider.

KNEELING HIP FLEXOR STRETCH



Start in a half kneeling position, with your non-kneeling leg out in front of you with your knee at a 90-degree angle. On the side with the leg out in front, place your hand on your hip. While you push your hips forward to deepen the stretch, raise the arm on the side of the hip being stretched overhead.

BRIDGE



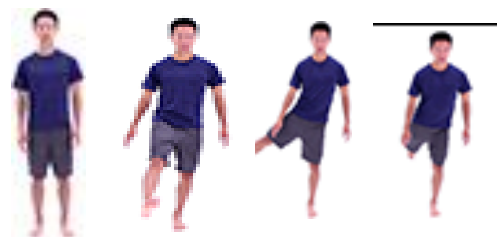
Begin on your back. Bend your knees and keep your back flat on the ground. Gently, contract your abdominals and hold to keep your back flat as you move up into the bridge position. Gently, squeeze your butt muscles and lift your hips off the ground.

SINGLE LEG STANCE



Start by standing up straight with a chair in front of you for balance. Raise one leg and maintain your balance as best as you can. Raise your opposite arm straight up by your ear and maintain your balance.

SINGLE LEG BALANCE CLOCK



Start by standing up straight and raising one foot slightly off the ground. Brace your abdominals and squeeze your butt cheek on the grounded leg. Slowly bring your raised, straight leg forward and back to neutral. Then, Bring it out to the side and back and then backward and back to

BRIDGE MARCHES



Begin on your back. Bend your knees and keep your back flat on the ground. Gently, squeeze your butt muscles and lift your hips off the ground. The goal is to have your hips aligned with your knees and shoulders. Hold this position and alternate lifting your feet off of the ground.

WIDE STANCE WALKING



Start by standing with a band wrapped around your ankles and your feet together. Bend slightly at your hips and knees and dig your heels into the ground. Slowly step one foot forward and out past your shoulder against the band. Bring your other foot up so your feet are together again.

FIRE HYDRANTS



Start on your hands and knees. Slowly raise one leg out to the side without straightening your knee and squeeze your buttock on that side. Keep your knee bent to 90-degrees throughout the movement so you only move at your hip. Slowly lower it back down.

CORRECTIVE EXERCISES FORWARD LUNGES

HIP FLEXOR STRETCH



Start by lying on a table/bed at a 45-degree angle so one whole leg is hanging off the table. Let your leg hang so that you feel a stretch in the front of your hip.

PIGEON STRETCH



Start by standing facing a table or counter. Raise your leg and place your lower leg on the edge of the table so it is perpendicular with you and parallel with the edge. Slowly lean forward over your leg until you feel a stretch in your hip on the raised side.

REVERSE LUNGE



Start by standing up straight with your hands behind your head. Keep your abdominals braced and your chest up as you step one foot back and bend both knees to about 90-degrees into a lunge. Drive through your front foot and step your back foot forward next to your front foot to stand up.

WALL SQUAT



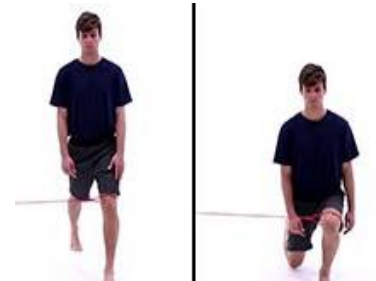
Start by standing with your back against a wall and your feet shoulder width apart. Slowly lower your body down the wall, digging your heels into the ground until your knees are at about a 90-degree angle. Then, press through your heels to raise your body back up.

GLUTEAL ISOMETRIC



Start by standing next to a wall so you are a few inches away on the side nearest to the wall. While standing upright, raise your working leg up so your hip and knee are bent to 90-degrees. Rotate your upper leg slightly outward so your knee presses into the wall.

MEDIAL STABILIZATION LUNGE



Start by standing up straight with a band anchored at knee height to your side. Step your leg farther from the band forward and loop the band around your front leg, just under your knee. Bend both knees to about 90-degrees to lower yourself into a lunge and then stand back up, keeping your legs staggered.

HIGH STEP UP



Start by standing with a Bosu ball in front of you with the curved side up. Step up onto the Bosu with one foot and then drive your other knee up toward the ceiling along with your opposite arm. Make sure to keep your abdominals contracted and your balance under control. Step back off and repeat on the other side.

BIRD DOG



Start on your hands and knees on a mat or table. Brace your abdominals and keep your back flat. Slowly raise one arm straight out in front of you while simultaneously raising your opposite leg back behind you until both limbs are straight and parallel with the ground. Lower them back down and repeat with your other limbs.

SINGLE LEG SQUAT



Stand facing away from a chair and raise one leg straight out in front of you. Slowly bend at your hip and knee to lower your butt down until it touches the edge of the chair. Then, press through your heel to stand back up. Make sure to keep your other leg raised off the ground and to keep your abdominals braced.

CHILD POSE WITH CROSS REACH



Begin on your hands and knees. Spread your knees wide apart while keeping your big toes touching. Sit on your heels. Bow forward placing your torso in between your thighs. Keep your arms extended and your palms facing up. Reach one arm straight under your torso and across your body as far as possible, keeping your palms up.

SIDE LYING WITH T SPINE ROTATION



Start by lying on your side with your knees bent and legs stacked on top of each other. Bring your arms straight out in front of you with your palms touching and your bottom palm facing up. Slowly raise your top arm straight up over your body to your opposite side. Rotate your torso and head along with your arm but do not let your legs move at all. Turn until your arm reaches the ground on your opposite side at shoulder height.

STANDING LAT STRETCH



Place your arms straight out in front of you so your hands are face down on the surface and step back slightly. Keep your lower and mid back flat and bend at your knees if necessary to bend forward and aim your chest toward the floor between your arms.

PRONE SCAPULAR RETRACTION "W"



Start by lying on your stomach on a mat with your forehead down on a towel. Place your arms out to your sides with your palms down and your elbows bent to about 90-degrees so your body makes a "W" shape with your arms. Slowly raise your arms straight up off the ground, and pulling your shoulder blades together.

PRONE SCAPULAR RETRACTION "T"



Lie on your stomach with your forehead down on a towel. Place your arms straight out to your sides with your palms down and your elbows straight so your body makes a "T" shape with your arms. Slowly raise your arms straight up off the ground, pulling your shoulder blades together.

PRONE SCAPULAR RETRACTION "I"



Start by lying on your stomach with both arms straightened up above your head. Hold a dumbbell in each hand so that the dumbbells are perpendicular to the floor. Pull your shoulder blades together back and down and lift your arms upwards slightly towards the ceiling. Slowly lower and relax your arms back to the starting position.

SHORT ARC ROTATION



Start by standing up straight. Place an exercise band in a door and wrap it around your back. Hold the band with both hands at the opposite shoulder. Face directly towards door. Rotate slowly backwards/forward through the upper trunk making a short arc rotation. Your hips should stay stable/still.

DIAGONAL EXTENSION "SWORD"



Start by anchoring a resistance band high up and standing sideways relative to the anchor. Grab the end of the band with your hand on the side closer to the anchor. Begin with your hand above your head and slightly to the side. Keeping your elbow straight, bring your arm down and across your body. When fully lowered, slowly return to starting position.

DIAGONAL CHOP



Stand with your feet just outside shoulder width apart and anchor the resistance band above you to your side. Place both hands on the free handle of the band and slowly, twist your torso, keeping your arms almost straight, and bring the band down diagonally across your body in a chopping motion.



CORRECTIVE EXERCISES: SHOULDER REACH

INTERNAL ROTATION WITH STRAP



Start by standing up straight with one end of a belt or strap in each hand. Hold your unaffected arm over your head and use it to help pull your affected arm behind your back by your hip by extending your overhead elbow.

DOORWAY PEC STRETCH



Start by standing in a doorway. Lift both arms until they are parallel to the floor and bend your elbows to 90 degrees. Place your forearms along the doorframe and let your body weight shift forward until you feel the stretch.

OVERHEAD TRICEP STRETCH



Start by standing up straight. Raise one arm and reach down your back behind your head with your elbow pointing straight up. Gently use your other hand to press your elbow down until you feel a stretch in the back of your raised arm.

WALL PUSH UP



Lean forward and place your hands on the wall, keeping your arms straight. Slowly brace your abdominals and bend your elbows to lower your body closer to the wall. Press back up with your hands, straightening your elbows back out.

BALL ON WALL STABILIZATION



Start by facing a wall and placing a small ball on the wall directly in front of you with your arm completely extended. Maintain a gentle push into the ball to hold it in place. Keeping your elbow straight, press the ball into a wall and gently move your arm up and down and then in small circles, both clockwise and counterclockwise.

SCAPULAR RETRACTION "Y"



Start by lying flat on your stomach on a mat or table. Place your arms so your body makes a "Y" on the mat and your palms are facing down. Pull your shoulder blades back and down and raise your arms straight up toward the ceiling. Slowly lower them back down.

PASS THROUGH



Start by standing up straight with one end of an exercise band in each hand. Spread your hands so that your arms are straight with your hands outside shoulder-width. Brace your abdominals and keep your back straight throughout the motion. Bring the band straight up over your head and behind your back so that the band hits near your butt. Bring the band straight back up over your head and in front of you.

BOW AND ARROW



Start by standing up straight with one end of an exercise band in each hand. Elevate one arm straight out in front of you and then pull the other end of the band back like you are drawing a bow. Make sure to keep your arms level and your shoulder back.

WALL WALKS



Start with a band looped around your arms just above your wrists. Place your forearms on the wall with your elbows bent to 90-degrees. Slide your forearms and elbows away from each other against the band along the wall slightly so you feel tension in your shoulders. Slowly slide one hand up the wall a few inches and then follow by sliding the other hand up the same distance. Continue up the wall one hand at a time as directed and then slide them back down one at a time.

HAMSTRING STRETCH



Place a belt around the foot of the leg you want to stretch and lay facing up. With your knee as straight as possible, lift your leg up and back holding the belt with both hands. Pull the belt to increase the stretch.

HIP FLEXOR STRETCH



Start by lying on a table at a 45-degree angle so one whole leg is hanging off the table. Let your leg hang so that you feel a stretch in the front of your hip.

PELVIC TILT



Start by lying on your back with your knees bent and your hands on your hips. Engage your abdominal muscles and then tilt your pelvis into the floor. Make sure the movement all comes from your hips and the rest of your body stays relatively still.

BRIDGE



Begin on your back. Bend your knees and keep your back flat on the ground. Gently, contract your abdominals and hold to keep your back flat as you move up into the bridge position. Remember to breathe. Gently, squeeze your butt muscles and lift your hips off the ground. The goal is to have your hips aligned with your knees and shoulders.

PLANK



Start by lying face down on the floor. Bring your elbows underneath your shoulders. Elevate your body off the ground. Your feet, hips and torso should all be in a straight line.

SUPERMAN



Start by lying flat on your stomach with your arms straight out in front of you and your legs straight out behind you. Raise your head and chest straight up along with one arm and your opposite leg.

ANTI ROTATION SQUAT



Start by standing up straight with an exercise band anchored to the side of you at about hip height. Hold the end of the band in both hands with your hands clasped by your chest and your elbows bent. Slowly extend your arms and press the end of the band straight out in front of your chest without letting it rotate you. Bend at your hips and knees to go into a squat with your arms straight out still and then stand back up straight.

ROTATION LUNGE



Stand up straight holding a small exercise ball. Step forward into a lunge by pressing into the floor with your heel on your front foot and your toe on your back foot. Lower your knee toward the ground and push the ball out in front of you. While in the low position, twist your body so the ball moves over your front leg. Return to the center. Repeat with your other leg forward.

DEAD BUG



Start by lying flat on your back. Brace your abdominals and raise your legs with your hips and knees bent to 90-degrees. Raise both arms straight over your chest. Slowly lower one arm back over your head and extend one leg straight out without touching the floor. Bring them back in and repeat with the opposite arm and leg.